Sentiment Analysis for Optimization on Investiment Strategies

- Stock Market News Sentiment Analysis
- Post Graduate Program in Generative AI for Business Applications
- Jaime Nagase
- 10/29/2025

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Contents / Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- EDA Results
- Data Preprocessing
- Model Performance Summary
- Appendix

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Executive Summary

- **The Problem:** Our company needs a faster way to understand stock market news and take a quick decisions. Today, our analysts must read many articles every day. This is slow, and the news can impacts stock prices. We are have an opportunity to improve the process and provide a better information fast enough for our analysts.
- **Our Project:** We built an Artificial Intelligence (AI) system to improve the process. This system reads news articles automatically, and then classifies the sentiment of each article as positive (1), neutral (0), or negative (-1). This will help our team understand market sentiment faster.
- **How We Did It:** We tested four different AI models to find the best one. We used two model types (Random Forest and Neural Network) and two methods to understand the text (Word2Vec and Sentence Transformers). We used all news before April 2019 to "teach" the models (train) and all news after that date to test their performance.
- **Key Results:** The best model was the Neural Network using Sentence Transformers. This model gave the most accurate results. It correctly classified 77.5% of the new (test) articles. The analysis shows this model is very good at identifying positive and negative news.
- **Our Recommendation:** We recommend using this model to support our financial analysts. It will give them fast and clear insights. With this tool, our team can make more informed investment decisions and get better results for our clients.

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Business Problem Overview and Solution Approach

Business Problem Overview: Develop an AI-driven sentiment analysis system to address the challenge of interpreting stock-related news articles to gauge market sentiment. It will:

- enhance the accuracy the stock market predictions
- optmize investment strategies
- analyse its impact on stock prices daily
- summarize the News at a Weekly level
- be specific on companies listed under NASDAQ
- focused on on daily stock prices and trade volumes
- empower financial analysts with actionable insights to lead a more informed investment decisions and outcomes for the customers

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Business Problem Overview and Solution Approach

Solution approach / methodology:

1 – Data loading covering news from xx/xx/xxxx to xx/xx/xxxx to compare information, and analyses the sentiment correlated to stock market prices metrics with open, mininum, high, and close price market of the NASDAQ stock market using EDA (Exploratory Data Analyis) using size of the texts to understand the complexity of them.

2 – Trasformers to convert text to numbers using two tecnics: a) NLP to translate news to vectors using Word2Vec, and b) Sentence Transformers based on BERT to understand the sentences using 384 dimensions for each news.

3 – Modeling and selection: we have combined 4 classification models: a) random forest + Sentence Transformer, b) Random Forest + Word2vec, c) Neural Network + Word2Vec, and d) Neural Network + Sentence Transformer. We compared all 4 models based on data test performance using acuracy and F1-Score as our primary metrics.

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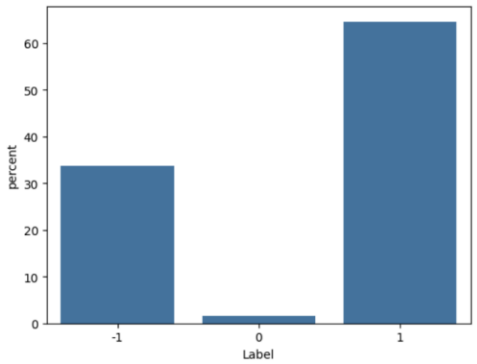
EDA Results

The distribution of the sentiment was unbalanced. Using countplot on label (graph below) we could see 35% (Label -1) as negative news, 3% (Label 0) as a neutral news, and finally 65% (Label 1) as a positive ones. It was unbalanced because having only 3% of neutral news.

```
sns.countplot(data=stock, x='Label', stat="percent");
```



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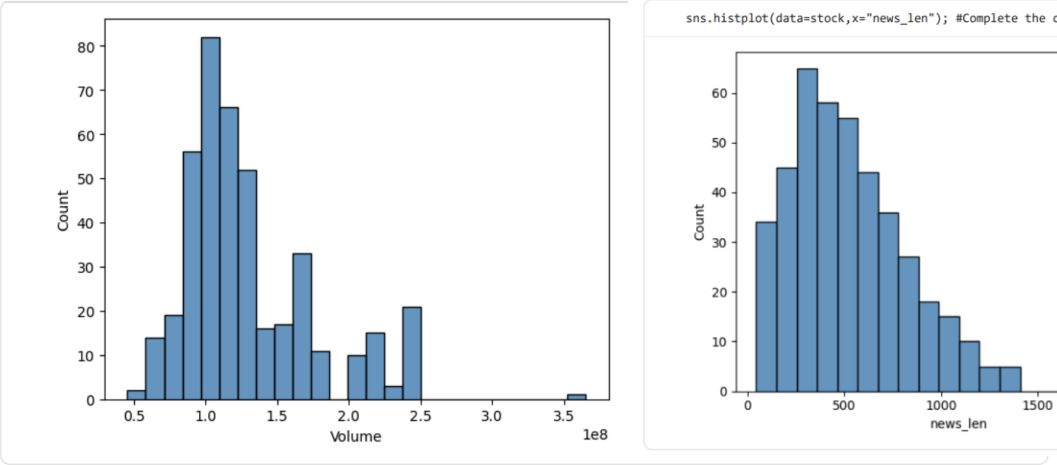
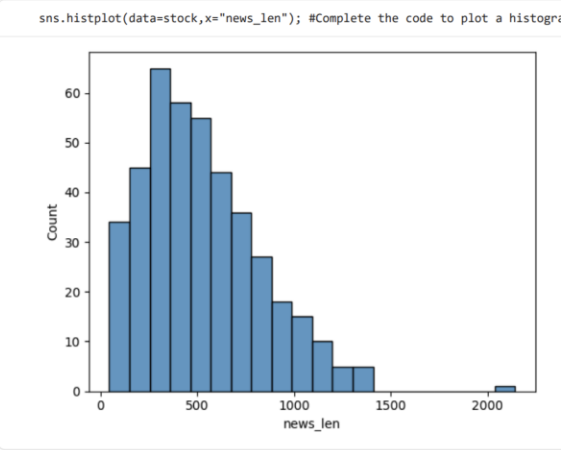
EDA Results

The size of each news has a huge range between them, which we must have use NLP advanced models to support this range, as we can see using histplot python function build the graphs below.




```
sns.histplot(data=stock,x="news_len"); #Complete the code to plot a histogra
```




```
#Calculating the total number of words present in the news content.
stock['news_len'] = stock['News'].apply(lambda x: len(x.split(' ')))

stock['news_len'].describe()    #Complete the code to print the statistical
```

	news_len
count	418.000000
mean	525.662679
std	303.584080
min	44.000000
25%	304.250000
50%	480.000000
75%	700.500000
max	2142.000000

dtype: float64

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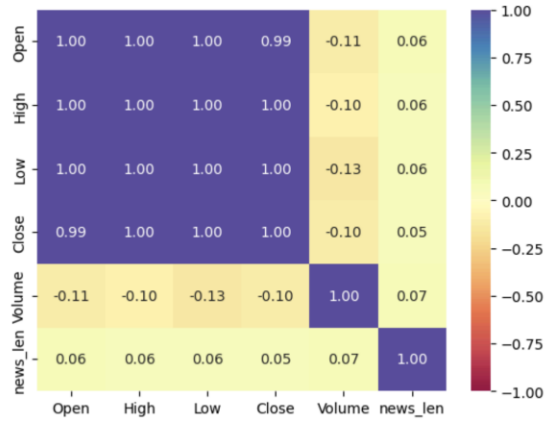
EDA Results

Correlation using bloxplots between news and the market stock price labeled by Volume, Close, Low, high, and open market stock price, where 1 is positive means positive news and –1 is means negative news

```
cols = ['Open','High','Low','Close','Volume','news_len']
sns.heatmap(
    stock[cols].corr(), annot=True, vmin=-1, vmax=1, fmt=".2f", cmap="Spectral"
)
plt.show()
```



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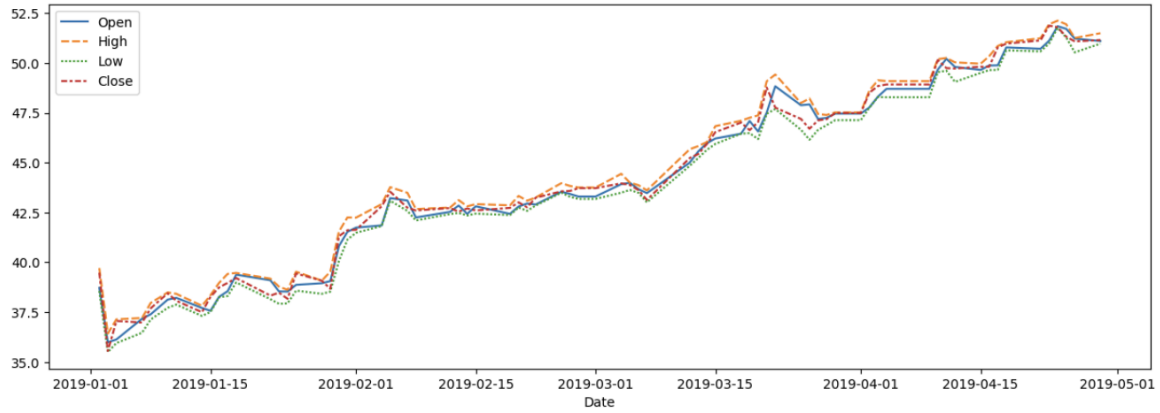
EDA Results

One concern raised after using lineplot with volume market prices, which is that all data was analyzed during a Bull Market, which means the data is based on a macroeconomic scenario where the market only increases.

```
plt.figure(figsize=(15,5))
sns.lineplot(stock_daily.drop("Volume", axis=1)); #Complete the code to plot a lineplot of all the variables except Volume
```



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Data Preprocessing

First we needed to make sure the data was accurated to be used in the models, and we checked if it has or not duplicated or null data in the dataset, using duplicated().sum function, and we did not find any issue.

Checking the duplicate values

```
stock.duplicated().sum() #Complete the code to check the duplicate values
```

```
0
```

Checking for missing values

```
stock.isnull().sum()
```

	0
Date	0
News	0
Open	0
High	0
Low	0
Close	0
Volume	0
Label	0

dtype: int64

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Data Preprocessing

First to guarantee that our data is reliable and we needed to verify it has duplicated or null data in the dataset, using duplicated().sum function. This way we make sure we do not have any duplicated news providing weight repeated or missing values that we need to fill them up.

Checking the duplicate values

```
stock.duplicated().sum() #Complete the code to check the duplicate
```

```
0
```

Checking for missing values

```
stock.isnull().sum()
```

	0
Date	0
News	0
Open	0
High	0
Low	0
Close	0
Volume	0
Label	0

dtype: int64

Data Preprocessing

```
stock['Date'].describe()
```

	Date
count	418
mean	2019-02-14 12:24:06.889952256
min	2019-01-02 00:00:00
25%	2019-01-11 00:00:00
50%	2019-01-31 00:00:00
75%	2019-03-21 00:00:00
max	2019-04-29 00:00:00

dtype: object

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Data Preprocessing

We used all news before 04/01/2019 to train the models (347 registers), and we have used after 04/01/2019 to test the models (71 registers). This way we can compare the trained model against a "future data" and see if the results were corrected.

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Sentiment Analysis - Model Evaluation Criterion

Classification Metrics: We have used Accuracy, Precision, Recall and F1-score metrics to evaluate the final performance of each data model against training data and test data through "model_performance_classification_sklearn" function. We calculated those metrics using "average=weighted", because our dataset was unbalanced (there we have much less Neutral news, than Positive, or Negative news).

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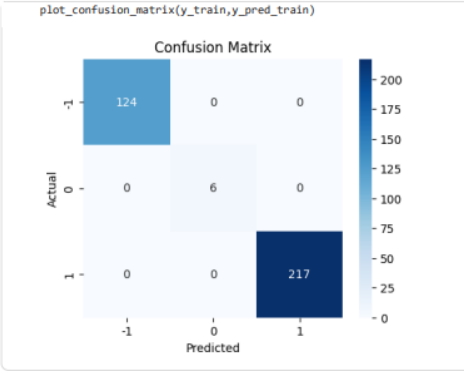
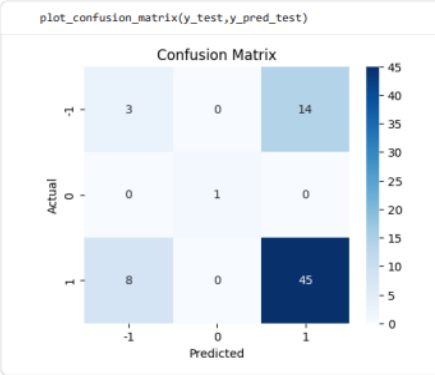
Sentiment Analysis - Model Evaluation Criterion

The Confusion Matrix provided a visual comparison of the models:

Random Forest + Word2Vec





We can see here the model had 69% of accuracy, which: a)14 negative news was classified as positive, b) just 1 neutral new was correct indicating a strong overfitting. And here we can say the model was not performing well with new data.

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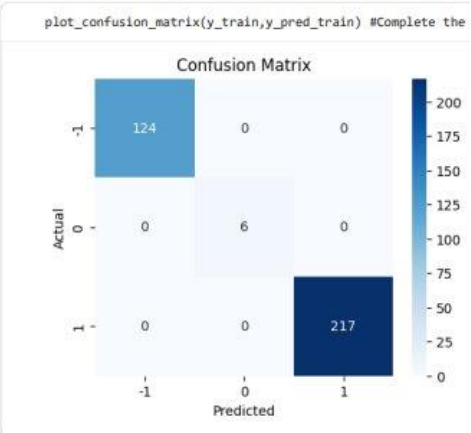
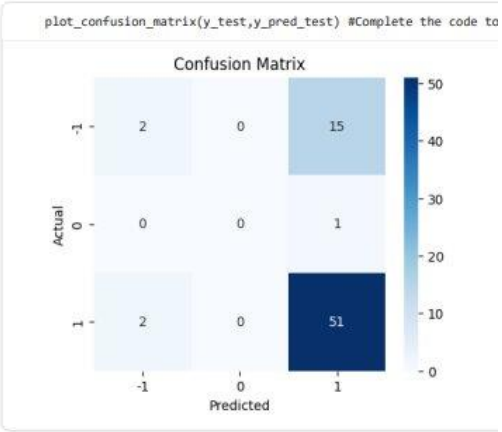
Sentiment Analysis - Model Evaluation Criterion

The Confusion Matrix provided a visual comparison of the models:

Random Forest + Sentence Transformer





We can see here the model had **74.6%** accuracy , which: a) is much more effective at identifying positive news (hitting 51) , but still misclassified **15** negative news as positive , b) **zero** neutral news were correct . This indicates strong overfitting, and we can say the model was still not performing well with new data.

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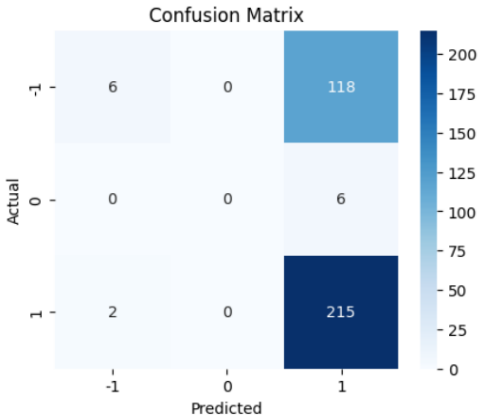
Sentiment Analysis - Model Evaluation Criterion

The Confusion Matrix provided a visual comparison of the models:

Neural Network + Word2Vec

```
plot_confusion_matrix(y_train,y_train_preds) #Complete the
```



We can see here a different problem, which this model is underfitting. The training accuracy was only 63.7%, and looking at the training matrix : a) the model misclassified 118 negative news as positive, b) it failed toidentify all neutral news. This shows the model failed to learn the patterns, even though the test accuracy was 76.1%. The model cannot tell the difference between negative and positive news.

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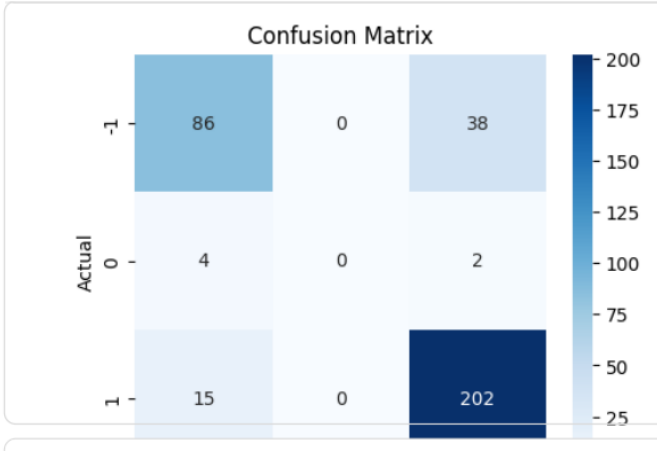
Sentiment Analysis - Model Evaluation Criterion

The Confusion Matrix provided a visual comparison of the models:

Neural Network + Sentence Transformer



The training accuracy was good achieving 83%. Different of the other models, it is not overfitting nor underfittting, because the test accuracy was also good at 77.5%. Looking at the matrix : a) it is very strong at finding positive (202) and negative (86) news , b) the main error is confusing 38 negative news as positive . This model learned the patterns and performs well on new data.

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Sentiment Analysis - Model Building & Evaluation

Overview of the models: We have combined 4 models to find the best solution, comparing two model architectures a) Random Forest, b) Neural Network with two NLP technologies 1) Word2Vec, and 2) Sentence Transformers.

Summary of key performance metrics:

Model	Embedding (NLP)	Accuracy (Train)	Accuracy (Test)	F1-Score (Test)
Random Forest	Word2Vec	100%	69%	66.50%
Random Forest	Sentence Transformer	100%	74.60%	68%
Neural Network	Word2Vec	63.70%	76.10%	67%
Neural Network	**Sentence Transformer**	**83%**	**77.50%**	**75.70%**

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Sentiment Analysis - Model Building & Evaluation

Comments or Observations

- We could conclude that Random Forest model is overfitting in this case, where it showed 100% of accuracy, but it was not performing the same with test dataset.
- Neural Network with Word2Vec was not good enough showing 63.7%, which indicates it was not able to understands the data pattern.
- Neural Network with Sentence Transformer was the best showing a good balance between training and test, 83% and 77.5% respectively, proving he learned and applied it in the test dataset.
- Sentence Transformer in all comparisons was better to understand context than word2vec.

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Sentiment Analysis - Model Building & Evaluation

The best model was **Neural Network with Sentence Transformer** because:

- Neural Network with Sentence Transformer was the best showing a good balance between training and test, 83% and 77.5% respectively, proving he learned and applied it in the test dataset.
- Among the all tests, this combination showed the best accuracy with 77.5% and the best F1-Score on test data set with 75.7%.
- It showed the best balance between learning and generalization, which means it not overfitted, showing it was reliable model.

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APPENDIX

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Data Background and Contents

Dataset provided by a Investments Startup: stock_news.csv, which contains:

- Daily news from NASDAQ
- 418 registers covering a period between 01/02/2019 to 04/29/2019

Data Dictionary

- Date: The date the news was released
- News: The content of news articles that could potentially affect the company's stock price
- Open: The stock price (in $) at the beginning of the day
- High: The highest stock price (in $) reached during the day
- Low: The lowest stock price (in $) reached during the day
- Close: The adjusted stock price (in $) at the end of the day
- Volume: The number of shares traded during the day
- Label: The sentiment polarity of the news content
    - 1: Positive
    - 0: Neutral
    - -1: Negative

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